



SAE
Collegiate
Club



RULEBOOK

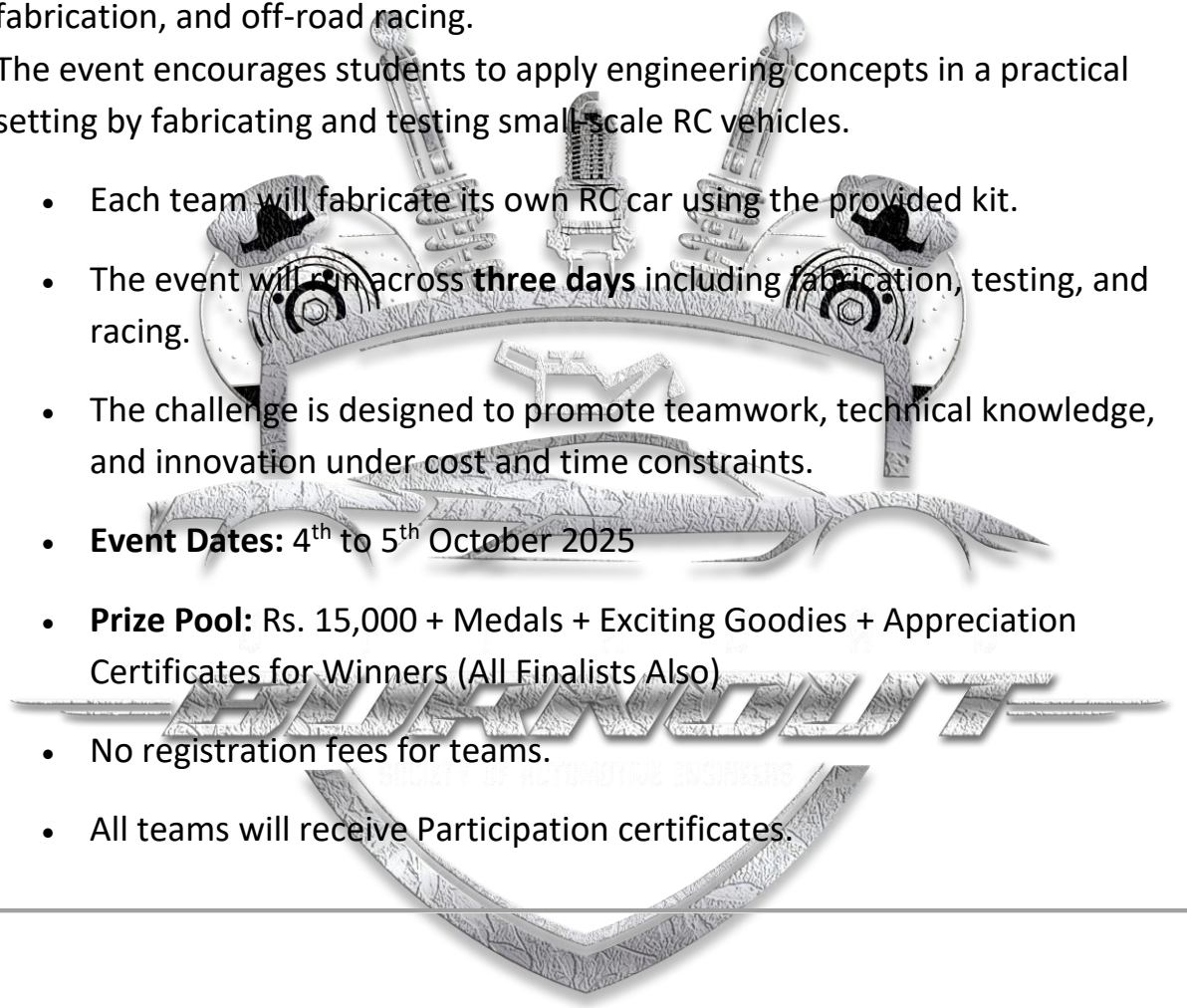
OFF ROAD BURNOUT – RULEBOOK

(SAE Collegiate Club, MMMUT Chapter)

1. Competition Overview

Off Road Burnout is an intercollegiate SAE event focused on RC car design, fabrication, and off-road racing.

The event encourages students to apply engineering concepts in a practical setting by fabricating and testing small-scale RC vehicles.

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- Each team will fabricate its own RC car using the provided kit.
 - The event will run across **three days** including fabrication, testing, and racing.
 - The challenge is designed to promote teamwork, technical knowledge, and innovation under cost and time constraints.
 - **Event Dates:** 4th to 5th October 2025
 - **Prize Pool:** Rs. 15,000 + Medals + Exciting Goodies + Appreciation Certificates for Winners (All Finalists Also)
 - No registration fees for teams.
 - All teams will receive Participation certificates.
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2. Participation Requirements

- Each team must consist of a **minimum of 2 and maximum of 5 members**.
- Mixed teams (minimum 2 boys & 2 girls) are encouraged with extra 10 points.

- Every team, irrespective of team size, must pay ₹3000 (minimal kit price).
 - Teams must adhere to timelines, safety norms, and instructions given by coordinators.
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3. Event Description

- **Orientation Day – Introduction & Material Distribution**
 - Theory and basic introduction about Mechanism.
 - Teams receive raw material and kit parts.
 - Presence of the Leader of each team is compulsory on the Orientation Day.
 - **Day 1 – Chassis Fabrication, Technical Inspection & Dynamic Rounds**
 - Chassis fabrication begins.
 - Teams complete full RC car assembly.
 - Preliminary technical inspection conducted.
 - Manoeuvrability, durability, traction tests.
 - **Day 2 – Pre-final race, Final race & Prize Distribution**
 - Head-to-head race among qualified teams.
 - Closing ceremony with results announcement.
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4. Dynamic Events & Scoring

The competition consists of **five dynamic stages**. The maximum total score achievable is **315 points**.

4.1 Technical Inspection (25 + 15 Bonus Points)

- RC car will be checked by event judges.
- Must comply with guidelines shared on Orientation Day.
- **25 points** for clearing inspection in first attempt.
- **20 points** for clearing inspection in second attempt.
- **+15 bonus points** for innovation/technology (within a \$1,000 budget).
- Each team must present valid GST bills of the innovation done in the car.
- Each team has **maximum 2 attempts**.

4.2 Manoeuvrability Test (50 Points)

- Teams must drive their RC car in an **“8-shaped” track**.
- **50 points** for successful completion in first attempt.
- **40 points** for successful completion in second attempt.
- Penalties:
 - **–5 points** if car leaves the track.
 - **–5 points** if car stops moving on track.
 - **–1 point per minute (Max 10 min time allowed)** if any car repair is required (valid only for second attempt)
 - After 10 minutes of car repairing, your team is disqualified.
- **2 attempts maximum** allowed.

4.3 Durability Test (50 or 25 Points)

- Cars must jump across a ramp.
- All 4 wheels must land successfully after the jump, and the car should be in working conditions without any physical damage.
- If there is any circuit failure after the jump, the car can be qualified after circuit maintenance.
- If team fails in their circuit maintenance, team will be disqualified.
- **50 points** – for first attempt.
- **35 points** – for second attempt.
- **2 attempts maximum** allowed.

4.4 Pre-final Race (100 to 35 Points)

- Teams race head-to-head on off-road track.
- 12 or 15 teams will be selected for the pre-final race, and the race will be in a group of 3 teams respectively.
- Teams will be qualified for the final race, on point-distribution basis.
 - **-5 points** if car leaves the track.
 - **-5 points** if car stops moving on track.
- Points distribution:
 - 1st Place – 100 pts
 - 2nd Place – 75 pts
 - 3rd Place – 55 pts
 - 4th Place – 35 pts

4.5 Final Race (75 to 25 Points)

- Further Qualified teams race head-to-head on off-road track.
- 4 or 5 teams which qualify the pre-final race, will be selected for the final race.
- No penalty of points in the final race.
- Points distribution: (if 4 teams are qualified for final race)
 - 1st Place – 75 pts
 - 2nd Place – 60 pts
 - 3rd Place – 45 pts
 - 4th Place – 30 pts
- Points distribution: (if 5 teams are qualified for final race)
 - 1st Place – 75 pts
 - 2nd Place – 60 pts
 - 3rd Place – 45 pts
 - 4th Place – 30 pts
 - 5th Place – 20 pts
- Finalists will be announced on points basis and all finalists will receive the Appreciation certificates.

5. Winner Declaration

- The **team with the highest cumulative score** from all rounds will be declared as the winner.
- In case of tie, race result will be considered as the winner deciding factor.
- Maximum achievable score = **325 points**.